**Unit:** Unit 3: Functions
**Topic:** Inverse Relations and Functions

**Name:** ____________
**Date:** __________

# Homework (Jalapeno)

**Q1.** The ancient Egyptians used a $y = x^2$ relationship to design pyramids. Which of the following is the inverse of this function?

(A) $y = \sqrt{x}$
(B) $y = -\sqrt{x}$
(C) $y = \sqrt{-x}$
(D) $y = -\sqrt{-x}$
(E) No inverse exists

**Q2.** The ancient Greek mathematician Euclid used the concept of inverse functions to calculate the area of a triangle. If $f(x) = 2x + 1$, what is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = \frac{x-1}{2}$
(B) $f^{-1}(x) = \frac{x+1}{2}$
(C) $f^{-1}(x) = 2x - 1$
(D) $f^{-1}(x) = \frac{x}{2} + 1$
(E) $f^{-1}(x) = \frac{x}{2} - 1$

**Q3.** The ancient Babylonians used a $y = x^3$ relationship to calculate the volume of a cube. Which of the following is the inverse of this function?

(A) $y = \sqrt[3]{x}$
(B) $y = -\sqrt[3]{x}$
(C) $y = \sqrt[3]{-x}$
(D) $y = -\sqrt[3]{-x}$
(E) $y = x^2$

**Q4.** The ancient Chinese mathematician Zu Chongzhi used the concept of inverse functions to calculate the area of a circle. If $f(x) = 3x - 2$, what is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = \frac{x+2}{3}$
(B) $f^{-1}(x) = \frac{x-2}{3}$
(C) $f^{-1}(x) = 3x + 2$
(D) $f^{-1}(x) = \frac{x}{3} + 2$
(E) $f^{-1}(x) = \frac{x}{3} - 2$

**Q5.** The ancient Mayans used a $y = 2^x$ relationship to calculate the population growth of a city. Which of the following is the inverse of this function?

(A) $y = \log_2 x$
(B) $y = \log_{1/2} x$
(C) $y = \log_2 x$
(D) $y = 2^x$
(E) $y = x^2$

**Q6.** If $f(x) = x^2 - 4$, which of the following statements is true about the inverse function $f^{-1}(x)$?

(A) The inverse function $f^{-1}(x)$ exists for all real numbers.
(B) The inverse function $f^{-1}(x)$ exists, but it is not a function.
(C) The inverse function $f^{-1}(x)$ does not exist.
(D) The inverse function $f^{-1}(x)$ is a linear function.
(E) The inverse function $f^{-1}(x)$ is a quadratic function.

**Q7.** If $f(x) = 2x + 5$ and $g(x) = \frac{x-5}{2}$, which of the following statements is true about the composition of $f$ and $g$?

(A) $f(g(x)) = x$
(B) $g(f(x)) = x$
(C) $f(g(x)) = 2x + 5$
(D) $g(f(x)) = \frac{x-5}{2}$
(E) $f(g(x)) = x - 5$

**Q8.** The ancient Egyptians used the concept of inverse functions to calculate the height of a pyramid. If $f(x) = x^2 + 2$, which of the following is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = \sqrt{x-2}$
(B) $f^{-1}(x) = -\sqrt{x-2}$
(C) $f^{-1}(x) = \sqrt{x+2}$
(D) $f^{-1}(x) = -\sqrt{x+2}$
(E) $f^{-1}(x) = x^2 - 2$

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## Open Ended Questions (FRQ)

**Q9.** The ancient Greek mathematician Archimedes used the concept of inverse functions to calculate the area of a circle. If $f(x) = 3x - 2$, find the inverse function $f^{-1}(x)$ and verify that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$. Show all work.

**Q10.** The ancient Chinese mathematician Liu Hui used the concept of inverse functions to calculate the volume of a sphere. If $f(x) = 2x^2 + 1$, find the inverse function $f^{-1}(x)$ and interpret its meaning in the context of the volume of a sphere. Show all work and explain your answer.

**Chef's Tips**

- Chef's Tips: Remind students to check their work carefully and to show all calculations.
- Chef's Tips: Emphasize the importance of understanding the concept of inverse functions and its applications.
- Chef's Tips: Encourage students to use graphing calculators or other technology to visualize the relationships between functions and their inverses.